



WeChat Enterprise

WeChat just launched WeChat Enterprise which is targeting the office going section. <http://dgit.in/WeChatEnterprise>



Hacked in 60 minutes

iPhone was hacked by a security firm in 60 minutes. How safe are our phones? <http://dgit.in/60minhack>

HOW WE TESTED

Last December we'd conducted our annual Zero1 awards and the router that won the award was the D-Link DIR-890L, a beast of a router with excellent transmission characteristics. Needless to say we used it for this benchmark. This is how we set up our testing infrastructure.

Since we wanted to test the repeater's capability, it was essential to keep both the transmission and reception wireless. So two end points were configured. One was our test rig which was connected to a PCIe Wi-Fi adapter with a massive 1900 Mbps bandwidth with 600 Mbps on the 2.4 GHz band. On the other end we used the D-Link DIR-890L which too has 600 Mbps on the 2.4 GHz band. A laptop with a Gigabit Ethernet port was connected using a CAT6 cable to the DIR-890L to ensure no bottlenecks. The repeater was placed at an equal distance from the two endpoints and the distance was increased to check transmission characteristics in different zones.

Zone 2 – Median – Roughly 30 feet

This is the ideal distance for most our test since signal strength had dropped to roughly 50 per cent at this range.

Zone 3 – Dead zone – Roughly 40 feet

This zone was behind 3 walls with a lot of high voltage cabling running between them, making it sort of a non-permeable surface. Only one repeater managed to pass this test.

Endpoint 1 - Test Rig

- CPU - Intel Core i7 6700K
- Motherboard - ASRock Z170 Extreme7+
- RAM - Corsair Vengeance LPX 32 GB
- SSD - SanDisk Extreme II 240 GB
- GPU - GALAX GTX 980 Ti
- Cooler - CoolerMaster TX3
- PSU - Corsair RM1000i
- PCIe - AC1900 Wi-Fi Adapter

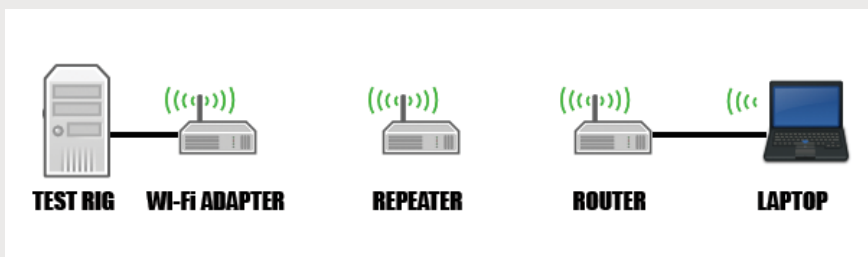
properties of the Wi-Fi adapter and Ethernet ports to ensure that bottlenecks would not come into the picture. Power saving features under the Green Ethernet Technology (802.3az) were turned off to ensure maximum performance. On the wireless front, any similar features were turned off. The performance difference was negligible but we decided to play it safe and keep them turned off.

Each repeater was then prepared for testing, the latest firmware was downloaded from the respective regional websites and updated. After the firmware update, the devices were reset and then finally set up in Extender mode. Antennae, if present, were aligned according to the guidelines mentioned in the manual or tilted with an angle of 45 degrees between them.

For benchmarking throughput, we used iperf3, the most versatile piece of software for measuring networking performance. TCP throughput was measured over a period of 20 seconds with a 40 KiloByte TCP window and 20 parallel streams. The same test was run five times with a time gap between runs to ensure that anomalies wouldn't creep into the test scores. Both TCP uplink and download bandwidth were obtained using this method.

We also used iperf3 for testing UDP transmission and calculating jitter. No window was set for UDP so as to allow the datagrams to be received without any hindrances. For short distances i.e. Zone 1 and 2 we used a datagram bandwidth of 1000 Mbit/s and 100 Mbit/s for Zone 3 since all higher bandwidth commands failed in Zone 3.

Lastly, we used WirelessMon to constantly monitor signal strength across all Zones.



Test Configuration

Zone 1 – Closest – Roughly 15 feet

The repeater was first setup over here and peak transmission rates were obtained when the repeater was closest to the endpoints.

Endpoint 2 - Laptop

- Router - D-Link DIR-890L AC3200
 - Connected via GbE to laptop
- On both end-points we modified the network

of -60 dBm. There should be a hemispherical band which falls in this particular signal strength. Figure out the best (often the centermost) spot and place your repeater. Most units come with WPS (Wi-Fi Protected Setup) so setting up is absolutely easy, just press the WPS button on the repeater and then press the WPS button on the router within a minute. The window for pressing the button could be as short as

20 seconds or as high as 180 seconds, so the sooner you can press it, the better it'll be. Most repeaters have an Ethernet port included so you can always manually set it up.

Note: Often repeaters will provide a randomly generated URL should you ever need to configure it later on. Do not lose this or you'll have to manually key in every local IP address within the subnet which amounts to 256 different

IP addresses to locate the UI of the repeater. Needless to say, that is not something you want to do, possibly ever.

BUILD QUALITY

We liked the way the D-Link DAP-1620 is built, the antennae rotate in such a manner that you can easily adjust the antenna field to get coverage along the horizontal and vertical plane. The WPS button is easily reachable and the built-in signal

strength indicator is a welcome feature. The unit is quite large and easily noticeable, something that might not appeal to certain folk. NETGEAR WN3000RP and TP-Link 860RE are quite similar to the DAP-1620, in that they both have external antennae but there are unique features to both as well. The 860RE has an AC mains pass-through socket which means you don't lose functionality of your AC mains socket just